

```
In [1]: thistuple = ("apple", "banana", "mango")
        print(thistuple)

('apple', 'banana', 'mango')
```

```
In [2]: thistuple = ("apple", "banana", "mango", "apple", "mango")
        print(thistuple)

('apple', 'banana', 'mango', 'apple', 'mango')
```

```
In [3]: thistuple = ("apple", "banana", "mango")
        print(len(thistuple))

3
```

```
In [4]: thistuple = ("apple",)
        print(type(thistuple))

        thistuple = ("apple")
        print(type(thistuple))

<class 'tuple'>
<class 'str'>
```

```
In [5]: thistuple = ()
        print(type(thistuple))

<class 'tuple'>
```

```
In [6]: tuple1 = ("apple", "mango", "cherry")
tuple2 = (1, 2, 3, 4, 5,)
tuple3 = (True, False, False)

print(tuple1)
print(tuple2)
print(tuple3)

('apple', 'mango', 'cherry')
(1, 2, 3, 4, 5)
(True, False, False)
```

```
In [7]: tuple1 = ("abc", 20, True)
print(tuple1)

('abc', 20, True)
```

```
In [10]: # Data Types
```

```
In [11]: mytuple = ("apple", "mango", "cherry")
print(type(mytuple))

<class 'tuple'>
```

```
In [12]: mytuple = tuple(("apple", "mango", "cherry"))
print(mytuple)

('apple', 'mango', 'cherry')
```

```
In [13]: mytuple = ("apple", "mango", "cherry")
print(mytuple[1])

mango
```

```
In [ ]: # Range
```

```
In [14]: mytuple = ("apple", "mango", "cherry")  
print(mytuple[-2])
```

mango

```
In [15]: mytuple = ("apple", "mango", "cherry", "banana", "orange", "kiwi")  
print(mytuple[2:4])
```

('cherry', 'banana')

```
In [16]: mytuple = ("apple", "mango", "cherry", "banana", "orange", "kiwi")  
print(mytuple[:5])
```

('apple', 'mango', 'cherry', 'banana', 'orange')

```
In [17]: mytuple = ("apple", "mango", "cherry", "banana", "orange", "kiwi")  
print(mytuple[2:])
```

('cherry', 'banana', 'orange', 'kiwi')

```
In [18]: # Negative Range
```

```
In [19]: mytuple = ("apple", "mango", "cherry", "banana", "orange", "kiwi")  
print(mytuple[-4:-1])
```

('cherry', 'banana', 'orange')

```
In [23]: mytuple = ("apple", "mango", "cherry")  
if "mango" in mytuple:  
    print("Yes, 'mango' is in the fruits tuple")
```

Yes, 'mango' is in the fruits tuple

```
In [24]: #Change Tuple Values
```

```
In [26]: x = ("apple", "mango", "cherry")
y = list(x)
y[1] = "kiwi"
x = tuple(y)

print(x)
```

('apple', 'kiwi', 'cherry')

```
In [29]: mytuple = ("apple", "mango", "cherry")
y = list(mytuple)
y.append("orange")
mytuple = tuple(y)

print(mytuple)
```

('apple', 'mango', 'cherry', 'orange')

```
In [30]: mytuple = ("apple", "mango", "cherry")
y = ("orange",)
mytuple += y

print(mytuple)
```

('apple', 'mango', 'cherry', 'orange')

```
In [31]: mytuple = ("apple", "mango", "cherry")
y = list(mytuple)
y.remove("apple")
mytuple = tuple(y)

print(mytuple)
```

('mango', 'cherry')

```
In [33]: mytuple = ("apple", "mango", "cherry")
del mytuple
print(mytuple) #this will raise an error because the tuple no longer exists
```

```
-----
NameError                                Traceback (most recent call last)
<ipython-input-33-99440dd0c858> in <module>
      1 mytuple = ("apple", "mango", "cherry")
      2 del mytuple
----> 3 print(mytuple) #this will raise an error because the tuple no longer exists

NameError: name 'mytuple' is not defined
```

```
In [35]: mytuple = ("apple", "mango", "cherry")
del mytuple
```

```
In [36]: mytuple = ("apple", "mango", "cherry")
```

```
mylist = list(mytuple)
del mylist[1]

mytuple = tuple(mylist)

print(mytuple)
```

```
('apple', 'cherry')
```

```
In [ ]: # Unpack Tuples
```

```
In [37]: fruits = ("apple", "banana", "cherry")
```

```
print(fruits)
```

```
('apple', 'banana', 'cherry')
```

```
In [38]: fruits = ("apple", "banana", "cherry")

        (green, yellow, red) = fruits

        print(green)
        print(yellow)
        print(red)
```

```
apple
banana
cherry
```

```
In [39]: # Using Astereisk*
```

```
In [40]: fruits = ("apple", "banana", "cherry", "strawberry", "raspberry")

        (green, yellow, *red) = fruits

        print(green)
        print(yellow)
        print(red)
```

```
apple
banana
['cherry', 'strawberry', 'raspberry']
```

```
In [41]: fruits = ("apple", "banana", "cherry", "strawberry", "raspberry")

        (green, *trophic, red) = fruits

        print(green)
        print(trophic)
        print(red)
```

```
apple
['banana', 'cherry', 'strawberry']
raspberry
```

```
In [42]: mytuple = ("apple", "banana", "mango")
         for x in mytuple:
             print(x)
```

apple
banana
mango

```
In [46]: mytuple = ("apple", "banana", "mango")
         for i in range(len(mytuple)):
             print(mytuple[i])
```

apple
banana
mango

```
In [47]: mytuple = ("apple", "mango", "cherry")
         i = 0
         while i < len(mytuple):
             print(mytuple[i])
             i = i + 1
```

apple
mango
cherry

```
In [48]: # Join Tuples
```

```
In [49]: tuple1 = ("a", "b", "c")
         tuple2 = (1, 2, 3)

         tuple3 = tuple1 + tuple2
         print(tuple3)
```

('a', 'b', 'c', 1, 2, 3)

```
In [50]: fruits = ("apple", "mango", "cherry")  
         mytuple = fruits * 2  
         print(mytuple)  
  
('apple', 'mango', 'cherry', 'apple', 'mango', 'cherry')
```

```
In [ ]:
```